On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

Xiang Huang

July 2026

Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences (a_n), (b_n) with $d_n = \mathrm{lcm}(1, \ldots, n)^3$. We prove *unconditionally* that $G_n = e^{o(n)}$ for a set of positive integers of natural density 1. The key ingredient is a new bound $Z(p) = o(p)$ on the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$, derived from the Apéry recurrence via a gap polynomial argument. Computation for all 9,590 primes $5 \le p \le 10^5$ gives $Z(p) \le 10$ with mean ≈ 1. Under the stronger arithmetic hypothesis that $Z(p) = O(1)$ on average (Hypothesis Z), we prove $\log G_n = O(\sqrt{n})$ for all n.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5)\, u_n - n^3\, u_{n-1}, \quad n \ge 1, \tag{1}$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n-th Apéry number. Set $d_n = \mathrm{lcm}(1, \ldots, n)^3$; Apéry [2] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$. Write $P(n) = 34n^3 + 51n^2 + 27n + 5$ for the middle coefficient.

2 The Wronskian identity

Lemma 1. *For all $n \ge 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. □

Corollary 2. *For every prime $p \ge 5$,*

$$v_p(G_n) \le 3\lfloor \log_p n \rfloor. \tag{2}$$

Proof. G_n divides $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6 d_n / n^3$. For $p \ge 5$, $v_p(6) = 0$ and $v_p(d_n/n^3) = 3\lfloor \log_p n \rfloor - 3 v_p(n) \le 3\lfloor \log_p n \rfloor$. □

3 Small primes: unconditional bound

Proposition 3. $\displaystyle\sum_{p \le \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n})$.

Proof. By (2), $v_p(G_n) \log p \le 3 \log n$ for every prime $p \ge 5$. Primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. □

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [6]). *For every prime p and $n = \sum_{i=0}^{s} n_i p^i$ in base p,*

$$b_n \equiv \prod_{i=0}^{s} b_{n_i} \pmod{p}. \tag{3}$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [10] and McIntosh [9].

5 The zero-count function

Definition 5. For a prime $p \geq 5$, define $Z(p) = \#\{j \in \{0, \ldots, p-1\} : b_j \equiv 0 \pmod{p}\}$.

5.1 Unconditional bound

Lemma 6 (No consecutive zeros). *For every prime $p \geq 5$ and every $j \in \{0, \ldots, p-2\}$, b_j and b_{j+1} cannot both vanish modulo p.*

Proof. Suppose $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$ with $0 \leq m \leq p-2$. The recurrence (1) at $n = m$ gives $(m+1)^3 b_{m+1} = P(m)\, b_m - m^3\, b_{m-1}$, so $m^3 b_{m-1} \equiv 0 \pmod{p}$. Since $1 \leq m \leq p-2$ implies $p \nmid m$, we get $b_{m-1} \equiv 0$. Iterating gives $b_0 \equiv 0 \pmod{p}$, contradicting $b_0 = 1$. □

Lemma 7 (Gap polynomial). *Fix $h \geq 2$. Suppose $b_m \equiv 0 \pmod{p}$ and $m + h \leq p - 1$. Then*

$$b_{m+h} \equiv C_h(m)\, b_{m+1} \pmod{p}, \tag{4}$$

where $C_h(m)$ is a rational function of m whose numerator (after clearing the denominator $\prod_{j=1}^{h-1}(m+j+1)^3$) is a polynomial of degree exactly $3(h-1)$ with positive leading coefficient.

In particular, if $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$, then by Lemma 6 $b_{m+1} \not\equiv 0$, so $C_h(m) \equiv 0 \pmod{p}$, and m is constrained to at most $3(h-1)$ residue classes modulo p, provided C_h is not identically zero over $\mathbb{F}_p$.

Proof. We iterate the recurrence from the initial condition $(b_m, b_{m+1}) = (0, b_{m+1})$. At step $n = m+1$: $b_{m+2} = P(m+1)\, b_{m+1}/(m+2)^3$, since $b_m = 0$ kills the second term. At step $n = m+j$ for $j \geq 2$: b_{m+j+1} is a $\mathbb{Q}(m)$-linear combination of b_{m+j} and b_{m+j-1}, both already rational multiples of b_{m+1}. The leading coefficient of $P(n)$ is 34 and the subtracted term has leading coefficient 1, so at each step the degree in m increases by 3 and the leading coefficient remains positive. After $h-1$ steps the numerator has degree $3(h-1)$.

For $h = 2$: $C_2(m) = P(m+1)/(m+2)^3$, with numerator $P(m+1) = (2m+3)(17m^2+51m+39)$, a cubic with leading coefficient 34. □

Lemma 8 (Nonvanishing over $\mathbb{F}_p$). *For every prime $p \geq 7$ and every $2 \leq h \leq p$, the numerator of C_h is not the zero polynomial modulo p.*

Proof. Write N_h for the numerator of C_h (a polynomial of degree $3(h-1)$), satisfying $N_{h+1}(m) = P(m+h)\, N_h(m) - (m+h)^6\, N_{h-1}(m)$, with $N_1 = 1$ and $N_2 = P(m+1)$.

First evaluation. Equation (4) at $m = -1$ (where $b_{-1} = 0$, $b_0 = 1$) gives $C_h(-1) = b_{h-1}$, hence $N_h(-1) = b_{h-1} \cdot \big((h-1)!\big)^3$.

Second evaluation. Setting $m = -2$ in the recurrence gives $N_{h+1}(-2) = P(h-2)\, N_h(-2) - (h-2)^6\, N_{h-1}(-2)$, with $N_1(-2) = 1$ and $N_2(-2) = P(-1) = -5$. By induction using the Apéry recurrence at $n = h - 2$:

$$N_h(-2) \;=\; -5\, b_{h-2}\, \big((h-2)!\big)^3 \qquad (h \geq 2). \tag{5}$$

Conclusion. If $N_h \equiv 0 \pmod p$ as a polynomial ($2 \leq h \leq p$, $p \geq 7$), then $N_h(-1) \equiv N_h(-2) \equiv 0$, forcing $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod p$ (since $p \nmid 5$ and $p \nmid (h-1)!$, $(h-2)!$). These are consecutive zeros, contradicting Lemma 6. □

Proposition 9. *For every prime $p \geq 7$,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \tfrac{3}{2}(H-1)(H-2) + 1, \tag{6}$$

where H is any integer with $2 \leq H \leq p$. Optimizing $H = \lfloor (p/3)^{1/3} \rfloor + 1$ gives $Z(p) \leq \frac{3^{4/3}}{2}\, p^{2/3} + O(p^{1/3}) = O(p^{2/3})$ unconditionally for all primes. In particular, $Z(p) = o(p)$.

Proof. Partition $\{0, \ldots, p-1\}$ into $\lceil p/H \rceil$ consecutive blocks of length at most H. In each block, mark the first zero; all other zeros have gap $h \in \{2, \ldots, H-1\}$ to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap-h pair constrains m to at most $3(h-1)$ residue classes mod p (the denominator factors $(m+j+1)^3$ are nonzero since $0 \leq m+j+1 \leq p-1$). By Lemma 8, $C_h \not\equiv 0$ over $\mathbb{F}_p$, so the numerator roots are at most $3(h-1)$. Summing:

$$Z(p) \leq \left\lceil \frac{p}{H} \right\rceil + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \tfrac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real $H > 0$ occurs at $H = (p/3)^{1/3}$, giving the bound. For $p = 2, 3, 5$: $Z(p) = 0$ (since b_j for $j < 5$ is coprime to $2, 3, 5$). □

Remark 10. The leading coefficient of the numerator of C_h is $U_{h-1}(17)$, the Chebyshev polynomial of the second kind evaluated at 17, satisfying $\ell_1 = 1$, $\ell_{h+1} = 34\ell_h - \ell_{h-1}$. Let $N_h(m)$ denote the cleared numerator of C_h. The identity $P(-u-1) = -P(u)$ (evident from $P(n) = (2n+1)(17n^2 + 17n + 5)$) propagates through the recurrence to give $N_h(-m-h-1) = (-1)^{h-1}\, N_h(m)$, verified computationally for $h \leq 200$. For even h, N_h is an *odd* polynomial in $(m + (h+1)/2)$, forcing the structural factor $(2m+h+1)$. At $h = 2$: factor $(2m+3)$ in $P(m+1) = (2m+3)(17m^2+51m+39)$. The computational data (Section 9) suggests $Z(p) = O(1)$, far stronger than $O(p^{2/3})$.

Remark 11 (Content restriction). A stronger nonvanishing holds: if $p \geq 7$ divides the content of N_h (the gcd of all its coefficients), then $h \geq 2p$. Indeed, $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} \big((r-1)!\big)^3 \big((h-r)!\big)^3$ for $1 \leq r \leq h$ (by induction from the Apéry recurrence); choosing $r = 1, 2$ and using $p \nmid (h-1)!$, $(h-2)!$, 5 for $h \leq p$ recovers Lemma 8, while the range $p < h < 2p$ is excluded similarly via $r = h - p + 1$.

Remark 12 (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$\big|\mathrm{Res}(N_h, N_{h+1})\big| = \prod_{j=1}^{h-1} \big(j!^3\, b_j\big)^6. \tag{7}$$

This follows from the recursion $\mathrm{Res}(N_h, N_{h+1}) = N_h(-h)^6\ \mathrm{Res}(N_{h-1}, N_h)$ (since $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$), combined with $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1}((h-1)!)^3 b_{h-1}$ (the reflection identity and the first evaluation from Lemma 8). Verified for $h = 2, 3, 4, 5$; the cases $h = 2, 3, 4$ give $\mathrm{Res}(N_2, N_3) = -5^6$, $\mathrm{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$, $\mathrm{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$.

A key consequence: $p \mid \mathrm{Res}(N_h, N_{h+1})$ if and only if $p \mid b_j$ for some $1 \le j \le h-1$ (when $p > h$, so factorials are coprime to p). In particular, N_2 and N_3 have disjoint root sets modulo p for all $p \ge 7$ (since $5^6 \ne 0$).

Remark 13 (Sharpness of the bound). The exponent 2/3 in Proposition 9 is optimal for any argument using only the degree bounds $\deg N_h = 3(h-1)$. The exact extremal problem $\max\{\sum r_h : r_h \le 3(h-1),\ \sum h\, r_h \le p\}$ has value $\frac{3}{2} p^{2/3} + O(p^{1/3})$, with leading constant 3/2 achieved by filling gap sizes $h = 2, 3, \ldots, H$ to capacity and using the remaining space at $h = H+1$, where $H \sim p^{1/3}$. Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield $O(p^{1/2})$: an abstract polynomial model on $\sim p^{2/3}$ points can satisfy $\deg F_h \le 3(h-1)$ for all relevant h. Improving the exponent requires genuinely arithmetic input about the family (N_h).

Remark 14 (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll's condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m)\, N_h(m+1) - N_{h+1}(m)\, N_{h+1}(m+1) = -\prod_{j=2}^{h+1} (m+j)^6. \tag{8}$$

This is an identity in $\mathbb{Z}[m]$, verified for $h \le 7$. A consequence: for $m \not\equiv -j \pmod{p}$ $(2 \le j \le h+1)$, the pairs $(m, h+2)$ and $(m+1, h)$ cannot both lie in the zero relation $\mathcal{Z} = \{(m,h) : N_h(m) \equiv 0\}$, since $N_{h+1}(m)\, N_{h+1}(m+1)$ would equal a nonzero product of sixth powers. This discrete Toda-lattice structure provides rigidity for the zero set of the family (N_h) beyond individual degree bounds.

Remark 15 (Projective orbit reformulation). Let $B_n = n!^3\, b_n$ and D_n be the second solution of the renormalized recurrence $w_{n+1} = P(n)\, w_n - n^6\, w_{n-1}$ with $(D_0, D_1) = (0, 1)$. Their Casoratian is $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$, nonzero for $n < p$. Define the *projective Apéry orbit* $\pi\colon \{0, \ldots, p-1\} \to \mathbb{P}^1(\mathbb{F}_p)$ by $\pi_n = [B_n : D_n]$. For $0 \le m < m+h < p$, $N_h(m) \equiv 0 \pmod{p}$ if and only if $\pi_m = \pi_{m+h}$. Thus the zero-count $Z(p)$ equals the number of distinct collisions in the orbit π. The reflection symmetry $P(-1-n) = -P(n)$ implies $b_{p-1-n} \equiv b_n \pmod{p}$, making the orbit a palindrome: $\pi_n = \pi_{p-1-n}$. In particular, $Z(p)$ is even except at the non-ordinary primes $p \mid a_p(f)$ (at most two primes $p \le 10^5$).

Remark 16 (Galois structure). For each fixed h, the Galois group of the splitting field of N_h over $\mathbb{Q}$ is not the full symmetric group: the reflection $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ forces roots into pairs $\{r, -r-h-1\}$, and $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ (hyperoctahedral group) with $m = \lfloor 3(h-1)/2 \rfloor$ (confirmed exactly for $h \le 5$, $2 \le h \le 9$ by obstruction tests). By the Chebotarev density theorem, the average number of $\mathbb{F}_p$-roots of N_h is 1 for odd h and 2 for even h. This prediction is verified: for 262 primes in $[1000, 3000]$, the measured averages are $0.98, 2.03, 1.11, 2.07, 1.11, 1.88$ for $h = 3, \ldots, 8$.

5.2 Computational evidence

Proposition 17. *For all* 9,590 *primes* $5 \le p \le 10^5$*:*

1. $Z(p) \in \{0, 2, 4, 6, 8, 10\}$ *for all but two primes; the two exceptions* ($p = 11$ *and* $p = 3137$) *have* $Z(p) = 1$ *(non-ordinary primes, where* $p \mid a_p(f)$*).* $\max Z(p) = 10$ *at* $p = 88{,}609$.

2. *The exact histogram is:*

$Z(p)$	0	1	2	4	6	8	10
count	5820	2	2926	722	106	13	1
fraction	.607	.0002	.305	.075	.011	.001	.0001
$Poi(\frac{1}{2})$	.607	—	.303	.076	.013	.002	.0002

The mean $Z(p) = 0.990$*, stable across all ranges.*

3. *The symmetry* $b_j \equiv b_{p-1-j} \pmod p$ *holds for all tested primes: zeros come in palindromic pairs* $\{j, p-1-j\}$*, so* $Z(p)$ *is even (except at non-ordinary primes). The pair count* $Z(p)/2$ *fits* Poisson$(1/2)$ *to within sampling noise (table above): the* j*-th pair appears independently with probability* $\approx 1/(2p)$*.*

5.3 Hypothesis Z

Hypothesis 18 (Hypothesis Z)**.** *There exists an absolute constant* C *such that* $Z(p) \le C$ *for all primes* p*.*

Remark 19. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ has a fundamentally different structure. Stienstra–Beukers [11] showed that A_p is related to a *CM* modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod p$ for all $p \equiv 3 \pmod 4$ — a *positive density* of primes with guaranteed zeros. In contrast, the $\zeta(3)$ Apéry numbers are governed by the non-CM newform $f \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis Z should hold for the $\zeta(3)$ sequence.

Remark 20. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod p$ [4, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [7]. Thus Hypothesis Z contains a known open problem as a special case.

6 The denominator connection lemma

Lemma 21. *Let* $p \ge 7$ *be prime with* $n/2 < p \le n$*, and set* $r = n - p$*. Then* $v_p(G_n) = 0$ *if and only if* $p \nmid b_r$*.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \ldots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \ge 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2 (variation of parameters). From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^{n} \frac{6}{m^3\, b_m\, b_{m-1}}. \tag{9}$$

For $p > n/2$, the only $m \in \{1, \ldots, n\}$ with $p \mid m$ is $m = p$.

Step 3. By Theorem 4, $b_p \equiv b_0\, b_1 = 5 \pmod p$ (since p has base-p digits $(0, 1)$), and by the symmetry $b_j \equiv b_{p-1-j}$, $b_{p-1} \equiv b_0 = 1 \pmod p$. Since $p \ge 7$, both are nonzero mod p.

The $m = p$ term in (9) has $v_p = -3$; all other terms have $v_p \ge 0$. Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p\, b_{p-1}} \equiv \frac{6}{5} \pmod p.$$

Using $b_n \equiv 5\, b_r \pmod p$ (Theorem 4, since $n = 1 \cdot p + r$):

$$p^3\, a_n \equiv \tfrac{6}{5} \cdot 5\, b_r = 6\, b_r \pmod p.$$

Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. □

Verification. For all $n \le 300$ and primes $p \in (n/2, n]$ with $p \ge 7$: $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases (100%).

7 The unconditional density-one theorem

Theorem 22. *For a set of positive integers n of natural density 1, $\log G_n = O(\sqrt{n})$, and in particular $G_n = e^{o(n)}$.*

Proof. Fix N large and consider $n \in (N, 2N]$. Write $B(n) = \#\{\text{bad primes } p > \sqrt{n} \text{ with } v_p(G_n) \ge 1\}$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$ for every n. Each bad prime contributes $v_p(G_n)\log p \le 3\log n$, so $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$. It suffices to show $B(n) = o(n/\log n)$ for all but $o(N)$ values of $n \in (N, 2N]$.

For $p > \sqrt{n}$, n has two base-p digits: $n = qp + r$ with $1 \le q \le \sqrt{2N}$ and $0 \le r < p$. By Theorem 4, $p \mid b_n$ iff $p \mid b_q$ or $r \in \mathcal{Z}_p$, where $\mathcal{Z}_p = \{j < p : p \mid b_j\}$. A prime p can be bad for n only if one of these holds.

Lower-digit incidence. For fixed $p \in (\sqrt{N}, 2N]$, an interval of N consecutive integers contains at most $N \cdot Z(p)/p + Z(p)$ values of n whose residue $r = n \bmod p$ lies in $\mathcal{Z}_p$. Set $\rho_p = Z(p)/p$. The total lower-digit incidence is

$$R_N \le \sum_{\sqrt{N}<p\le 2N} \big(N\rho_p + Z(p)\big).$$

By Proposition 9, $\rho_p \to 0$. Since $\pi(2N) = O(N/\log N)$:

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \le 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence $R_N = o(N^2/\log N)$.

Leading-digit incidence. For fixed q, $p \mid b_q$ requires p to be a prime divisor of b_q, and $b_q \le (q+1)\cdot 64^q$ gives at most $O(q)$ such primes. For each pair (q,p), at most p integers in $(N, 2N]$ have $\lfloor n/p \rfloor = q$. Hence the total leading-digit incidence is

$$L_N \le \sum_{q=1}^{\lceil\sqrt{2N}\rceil} O(q)\cdot \frac{2N}{q+1} = O(N\sqrt{N}) = o(N^2/\log N).$$

Conclusion. By Markov's inequality applied to $\sum_{n\in(N,2N]} B(n) \le R_N + L_N = o(N^2/\log N)$: the number of $n \in (N, 2N]$ with $B(n) > n/\log^2 n$ is $o(N)$. A dyadic decomposition shows the exceptional set has density zero. For all remaining n, $\log G_n = O(\sqrt{n}) + 3B(n)\log n \le O(\sqrt{n}) + O(n/\log n) = o(n)$. $\square$

8 Counting bad primes: the conditional result

Theorem 23. *Assume Hypothesis Z. Then for a set of positive integers n of natural density 1, $\log G_n = O(\sqrt{n})$.*

Proof. As in Theorem 22, decompose $\log G_n = \sum_p v_p(G_n)\log p$. The small-prime contribution is $O(\sqrt{n})$ unconditionally.

For primes $p > \sqrt{n}$: $v_p(G_n) \le 3$ and p is bad only if some base-p digit of n lies in $\mathcal{Z}_p$. Under Hypothesis Z with pointwise $Z(p) \le C$, for $n \in (N, 2N]$ the total lower-digit incidence satisfies

$R_N \le \sum_p (N \cdot C/p + C) = NC \sum_{p>\sqrt{N}} 1/p + O(N/\log N) = O(N)$. The leading-digit incidence $L_N = O(N\sqrt{N})$ (as in Theorem 22). By Markov's inequality, the number of $n \in (N, 2N]$ with $B(n) > (\log n)^2$ is $O(N/(\log N)^2) = o(N)$. For all remaining n, $\log G_n = O(\sqrt{n}) + 3B(n)\log n \le O(\sqrt{n}) + O((\log n)^3) = O(\sqrt{n})$. □

Remark 24 (On the all-n conjecture). The all-n statement $\log G_n = O(\sqrt{n})$ for *every* n would require bounding the count of bad primes $B(n)$ pointwise rather than on average. For a specific n, the events $\{n \bmod p \in \mathcal{Z}_p\}$ across different primes are essentially independent with probability $\le C/p$ each, so $B(n) = O(1)$ should hold for all n by the same heuristic that predicts $Z(p) = O(1)$. Making this rigorous requires either a sieve inequality controlling the moving-residue incidence, or a sufficiently strong equidistribution hypothesis on the Frobenius elements of the Apéry family.

Corollary 25. *The conjecture $G_n = e^{o(n)}$ for all n is equivalent to $Z(p) = o(p)$ (i.e., $\mathcal{Z}_p$ has sublinear density): if $Z(p) \ge \varepsilon p$ for a positive proportion of primes, double-counting gives $\log G_n \ge c\varepsilon n$ for a positive proportion of n.*

9 Computational verification

We computed G_n exactly for $n = 1, \ldots, 500$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	2.30
51–100	0.108	0.44
101–200	0.086	0.32
201–300	0.048	0.17
301–400	0.038	0.13
401–500	0.024	0.09

A power-law fit gives $\log(G_n)/n \sim 4.5\, n^{-0.85}$, i.e., $\log G_n \sim 4.5\, n^{0.15}$, well within $O(\sqrt{n})$.

10 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [3] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base-p digits.

2. For the gap-2 polynomial $C_2(m) = P(m+1)$, the factorization $P(x) = (2x+1)(17x^2+17x+5)$ shows that gap-2 pairs can only occur at $m = (p-3)/2$ (when $2(m+1)+1 = 2m+3 \equiv 0$) or at roots of the quadratic $17x^2 + 17x + 5$ (discriminant -51; exists over $\mathbb{F}_p$ iff $\left(\frac{-51}{p}\right) = 1$). Empirically, all observed gap-2 pairs arise from the linear factor at $m = (p-3)/2$.

3. For $1 \le j \le p-2$, the Apéry number $b_j \bmod p$ admits a character-sum representation as a Greene [5] Gaussian hypergeometric ${}_4F_3$ over $\mathbb{F}_p$ with parameter $\chi = \omega^j$ (Teichmüller character). However, the Weil bound controls the archimedean size $|b_j|_\infty = O(p^{3/2})$ but does *not* bound $Z(p)$, since zeros of b_j modulo p are a p-adic event. The gap polynomial argument above is specifically adapted to this p-adic zero-count problem.

4. Malik–Straub [8] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings.

References

[1] S. Ahlgren and K. Ono, *A Gaussian hypergeometric series evaluation and Apéry number congruences*, J. reine angew. Math. **518** (2000), 187–212.

[2] R. Apéry, *Irrationalité de $\zeta(2)$ et $\zeta(3)$*, Astérisque **61** (1979), 11–13.

[3] F. Beukers, *Some congruences for the Apéry numbers*, J. Number Theory **21** (1985), 141–155.

[4] F. Beukers, *Another congruence for the Apéry numbers*, J. Number Theory **25** (1987), 201–210.

[5] J. Greene, *Hypergeometric functions over finite fields*, Trans. Amer. Math. Soc. **301** (1987), 77–101.

[6] I. Gessel, *Some congruences for Apéry numbers*, J. Number Theory **14** (1982), 362–368.

[7] S. Jin, W. Ma, and K. Ono, *A note on non-ordinary primes*, Proc. Amer. Math. Soc. **144** (2016), 4591–4597.

[8] A. Malik and A. Straub, *Divisibility properties of sporadic Apéry-like numbers*, Res. Number Theory **2** (2016), Art. 5.

[9] R. J. McIntosh, *A generalization of a congruential property of Lucas*, Amer. Math. Monthly **99** (1992), 231–238.

[10] Y. Mimura, *Congruence properties of Apéry numbers*, J. Number Theory **16** (1983), 138–146.

[11] J. Stienstra and F. Beukers, *On the Picard–Fuchs equation and the formal Brauer group of certain elliptic K3 surfaces*, Math. Ann. **271** (1985), 269–304.